

Algebra 1
Unit 13
Lesson 8 Practice

Name Answer Key
Date _____
Period _____

Calista is running a marathon. Based on the temperature on race day and her average marathon time, she calculated that she needs to consume 120 fluid ounces throughout the race. She wants to use both sports drinks and plain water, which she will grab from aid stations along the course. She can get sports drink in 6-ounce cups and water in 8-ounce cups.

1. $6s + 8w$ is a(n)

- ☐ function.
- ☐ equation.
- ★ ☒ expression.

2. The values 6 and 8 are

- ★ ☒ coefficients.
- ☐ terms.
- ☐ constants.

3. The monomials $6s$ and $8w$ are

- ☐ coefficients.
- ★ ☒ terms.
- ☐ constants.

4. $6s + 8w = 120$ is a(n)

- ☐ function.
- ★ ☒ equation.
- ☐ expression.

5. Determine the amount of water Calista needs if she consumes 8 cups of sports drink.

$$6(8) + 8w = 120$$

$$\begin{array}{r} 48 + 8w = 120 \\ -48 \quad -48 \end{array}$$

$$\frac{8w}{8} = \frac{72}{8}$$

$$w = 9$$

9 cups of water

A farmer has 1,000 feet of fencing to enclose his property, which borders a lake. The diagram shows the width, w , of the property and the lake that his property borders. The fencing is not used along the lake.

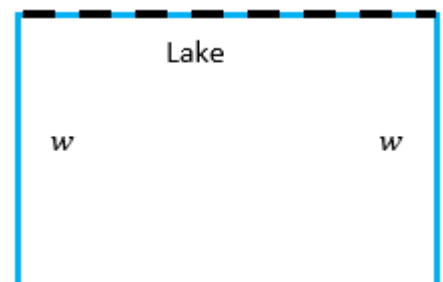
Explain what each quantity represents in this context.

6. $1000 - 2w$

The amount of fencing left after putting the fence on the left and right sides of the property. It is the length of the the property parallel to the lake.

7. $w(1000 - 2w)$

The area of the farmer's property.



Midori's parents started a college savings fund when she was born in 2010. The amount of money in the savings fund since 2010 can be represented by the equation $y = 10,000(1 + 0.085)^x$, where y is the amount in the college fund and x is the number of years since 2010.

8. The equation $y = 10,000(1 + 0.085)^x$ represents

- ☒ exponential growth.
☐ exponential decay.

9. The rate of

- ☐ decay
☒ growth

is 8.5 %.

10. What does the coefficient 10,000 represent?

The original amount of money deposited in 2010.